QIYAO WANG

Ph.D. Student @ Department of Mechanical Engineering, Tsinghua University, P.R. China
Tel: +86 15010819068 | Email: qy-wang21@mails.tsinghua.edu.cn | Homepage: King-Bridge.github.io

EDUCATION

Tsinghua University *2025.09 – present*
Ph.D. Student. in Mechanical Engineering
Adviser: Prof. Huichan Zhao
Research Interest: Soft Robot, Intelligent Control, Bio-inspired Design

Tsinghua University *2021.09 – 2025.07*
B.S. in Theoretical and Applied Mechanics & Aerospace Engineering
GPA: 3.85/4.00 (rank: 4/30)
Relevant Courses: Mathematical Analysis, Advanced Algebra, Continuum Mechanics, Fluid Mechanics, Pattern Recognition and Machine Learning, Signals and Systems, Automatic Control Theory
Language: TOEFL 103

PUBLICATION

[1] Weijie Qin†, **Qiyao Wang**†, Bingcen Gong†, Yijia Luo†, Seesoil: Mitigating Negative Emotions through a Desktop Robot for Oil Splash Incidents, *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, Student Design Challenge, 2026.
† These authors contributed equally to this work.

[2] Yinglei Zhu, Xuguang Dong, **Qiyao Wang**, Qi Shao, Fugui Xie, Xinjun Liu, Huichan Zhao*, EquiMus: Energy-Equivalent Dynamic Modeling and Simulation of Musculoskeletal Robots Driven by Linear Elastic Actuators, *Robotics and Automation Letter*, 2025.

[3] **Qiyao Wang**, Thileepan Stalin, Pablo Valdivia y Alvarado*, Architected Vacuum Driven Origami Structures via Direct Ink Writing of RTV Silicone, *Robotics and Automation Letter*, 2025.

[4] Yuhang Zheng†, **Qiyao Wang**†, Chengliang Zhong*, He Liang, Bo liang, Zhengxiao Han, and Yupeng Zheng, Enhancing Daily Life through an Interactive Desktop Robotics System※, *CAAI International Conference on Artificial Intelligence*, 2023.
† These authors contributed equally to this work.
※ Won the Best Demo Award in CICAI 2023.

SELECTED AWARDS AND HONORS

- **Excellent Graduate of Tsinghua University**, University-level honor (top 10%) *2025*
- **Disruptive Innovation Talent Fellowship**, Tsinghua University (for 40 undergrads) *2024*
- **Overall Excellence Scholarship**, Scholarship of Tsinghua University *2022, 2023, 2024*
- **1st Place**, ICRA Preparation and dish Up of an English Breakfast with Robots (PUB.R) competition *2023*
- **4th Place**, RoboCup 2023 Humanoid League – kid size *2023*
- **Technology Innovation Excellence Scholarship**, Scholarship of Tsinghua University *2023*
- **First Prize**, 6th CoreTech Winter Camp Smart Car Competition *2022*

RESEARCH EXPERIENCE

***1.* Dual MLP controller for pneumatic artificial muscle fiber bundles** *2023.10 – Present*
Advisor: Prof. Huichan Zhao, THU Soft Robotics Research Group, Department of Mechanical Engineering, Tsinghua University

This work aims to build and control a 7-dof multi-jointed soft robotic arm driven by multiple bundles of pneumatic artificial muscle (PAM). Positional control of single PAM fiber bundle should be accomplished first. To cope with the nonlinearity, hysteresis, and viscosity of PAM, this work designed a dual MLP controller and verified its control effect through experiments.

***2.* Architected Vacuum Driven Origami Structures via Direct Ink Writing** *2024.07 – 2024.09*
Advisor: Prof. Pablo Valdivia y Alvarado, Bio-inspired Robotics and Design Lab, Singapore University of Technology and Design, Singapore

This work uses Direct Ink Writing (DIW) and freeform liquid 3D printing (FL3DP) technologies with RTV silicone to manufacture soft origami structures. We apply an architected approach to design a library of flexible

hinges actuated by negative pressure , and developed a custom Python-based software tool to automatically generate 3D-printable G-code from user-defined geometric parameters.
This work produced a first-author paper in RAL.

3. **Research on organizing drawer interactively by a desktop robotics system** *2023.05 – 2023.07*

Advisor: Prof. Guyue Zhou, Discover Lab, Institute for AI Industry Research, Tsinghua University

This work uses a collaborative robotic arm with a Realsense d455 camera and a Robotiq two-finger gripper as the hardware. We apply Yolov7, Affordance and 3D Implicit Transporter for object detection, MoveIt for motion planning, and ChatGPT as a voice interface. We finally realized the job of making the robotic arm put the specified items into or take out from the specified drawer through natural language voice interaction.

This work produced a demo paper in CICAI 2023, and won the Best Demo Award in the conference.

4. **Design and control of breakfast-making robotic system** *2023.01 – 2023.06*

Advisor: Prof. Guyue Zhou, Discover Lab, Institute for AI Industry Research, Tsinghua University

The hardware for this work is a collaborative robotic arm with a Realsense d455 camera and a Robotiq two-finger gripper, with a movable chassis and LIDAR on it. We apply Yolov7 for object detection, MoveIt for motion planning, GraspNet for assisting grasping, and ChatGPT as voice interaction interface. We finally realized the cooking, serving and presentation of several English breakfast dishes, including hot tomatoes, grilled toast, fried eggs, bean dip, etc.

This work won the 1st Place of the PUB.R competition in ICRA 2023.

5. **Research on vision and navigation of humanoid robot soccer player** *2022.10 – 2023.07*

Advisor: Dr. Li Liu, Department of Mechanical Engineering, Tsinghua University

This work aims to build a humanoid robot soccer team, and focuses on the vision and navigation work of the robots. We apply Yolo-Fast to detect the ball, the goal and some iconic positions on the field. We then utilize some computer vision methods to accomplish distance measurement of the targets using a monocular camera. For navigation, we combined the detection results with the data from the IMU sensor, to localize our robots and navigate them to the target location.

This work won the 4th Place RoboCup 2023 Humanoid League.

ADDITIONAL INFORMATION

Sports and Arts

- Main player of Xingjian College soccer team, and helped the team win the second place in the Tsinghua University "John Ma" Soccer Cup - Grade B. *2021.10 - Present*
- Principal Clarinet player in the Tsinghua University Student Symphony Orchestra (TUSO), participated in many performances, and won the 1st place in the Beijing University Students' Art Exhibition with the team. *2021.08 - Present*

Volunteer Services

- Xingjian College – "Forerunner" Program: Help freshmen adapt to college life and find the right direction for themselves through communication and Q&A. *2022.09 – 2023.06*
- Tsinghua University Admission Volunteer: Introducing various college majors to high school graduates and assisting them in applying for a major. *2022.06*
- Tsinghua University – "Alma Mater" Program: Introducing my college and department to my high school underclassmen. *2022.02*